UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 009 · Particular integral when X is any other form · compiled 16 October 2026

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1 Short notes

1 · The general method: linear factors, partial fractions and $\frac{1}{D-m}X$
When X is not e^{ax}, $\sin/\cos(ax+b)$, x^m or $e^{ax}V$: CF as usual; factorise $f(D)=(D-m_1)\cdots(D-m_n)$; partial fractions $\frac{1}{f(D)}=\sum\frac{A_k}{D-m_k}$; each term by $\frac{1}{D-m}X=e^{mx}\int Xe^{-mx}dx$ (no constants). Or apply the factors one after another. $m=0$: integrate. Example: $D^2y=\ln x$: $y=c_1+c_2x+\frac{x^2\ln x}{2}-\frac{3x^2}{4}$.

2 · Complex linear factors: $(D^2+a^2)y=\tan ax$
$(D^2+a^2)y=\tan ax$: $\frac{1}{D^2+a^2}=\frac{1}{2ia}\left[\frac{1}{D-ia}-\frac{1}{D+ia}\right]$; $\frac{1}{D-ia}\tan ax=e^{iax}\int\tan ax\,e^{-iax}dx=-\frac{1}{a}[1+ie^{iax}\log(\sec ax+\tan ax)]$; $i\to -i$ for the other. PI $=-\frac{1}{a^2}\cos ax\log(\sec ax+\tan ax)$. tan, sec, csc can also be done by variation of parameters (class 011).

3 · Right-hand sides in e^x: the substitution $t=e^x$
X in e^x: $t=e^x$, $e^{kx}dx=t^{k-1}dt$. ESA Dec-23 3c: $\frac{e^{e^x}}{(D+1)(D+2)}=e^{-x}e^{e^x}-e^{-2x}(e^x-1)e^{e^x}=e^{-2x}e^{e^x}$, $y=c_1e^{-x}+c_2e^{-2x}+e^{-2x}e^{e^x}$. $\frac{\sin(e^x)}{(D+1)(D+2)}=-e^{-2x}\sin(e^x)$. $\frac{e^{2x}\sin(e^x)}{(D+2)(D+3)}=-\sin(e^x)-6e^{-x}\cos(e^x)+18e^{-2x}\sin(e^x)+24e^{-3x}\cos(e^x)$.

2 Keyword index

17 terms; the number in brackets is the topic.

$\frac{1}{D^2}$ (1). Integrate twice, no constants.

$\frac{1}{D^2}\ln x$ (1). $\frac{x^2\ln x}{2}-\frac{3x^2}{4}$.

$\sin(e^x)$ (3). PI $-e^{-2x}\sin(e^x)$ for $(D+1)(D+2)$.

e^{e^x} (3). ESA Dec-23 3c; PI $e^{-2x}e^{e^x}$ for $(D+1)(D+2)$.

$i\to -i$ (2). Gives $\frac{1}{D+ia}X$ from $\frac{1}{D-ia}X$ (real X).

$t=e^x$ (3). $dt=e^x dx$, $e^{kx}dx=t^{k-1}dt$.

Any other form (1). X not e^{ax}, $\sin/\cos$, x^m or $e^{ax}V$.

Bernoulli's rule (3). Repeated integration by parts in one line.

Complex factors (2). $D^2+a^2=(D+ia)(D-ia)$.

Euler's formula (2). $e^{\pm iax}=\cos ax\pm i\sin ax$.

Integrating factor (1). e^{-mx} for $u'-mu=X$ (why E1.1 holds).

Linear factors (1). $f(D)=(D-m_1)\cdots(D-m_n)$.

Partial fractions (1). $\frac{1}{f(D)}=\sum\frac{A_k}{D-m_k}$.

PI of $\tan ax$ (2). $-\frac{1}{a^2}\cos ax\log(\sec ax+\tan ax)$.

Standard formula (1). $\frac{1}{D-m}X=e^{mx}\int Xe^{-mx}dx$.

Successive application (1). $\frac{1}{D-m_1}\left[\frac{1}{D-m_2}X\right]$, no partial fractions.

Variation of parameters (2). Alternative for $\tan$, $\sec$, $\csc$ (class 011).

3 Rapid recall

1	When is this method used?	When X is none of e^{ax}, $\sin/\cos(ax+b)$, x^m, $e^{ax}V$.
2	$\frac{1}{D-m}X$?	$e^{mx}\int Xe^{-mx}dx$.
3	$\frac{1}{D+2}X$?	$e^{-2x}\int Xe^{2x}dx$.
4	Why does $\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}dx$ hold?	$u' - mu = X$; integrating factor e^{-mx}.
5	Partial fractions of $\frac{1}{(D+1)(D+2)}$?	$\frac{1}{D+1} - \frac{1}{D+2}$.
6	Partial fractions of $\frac{1}{D^2+a^2}$?	$\frac{1}{2ia}\left[\frac{1}{D-ia} - \frac{1}{D+ia}\right]$.
7	$\frac{1}{D^2}\ln x$?	$\frac{x^2\ln x}{2} - \frac{3x^2}{4}$.
8	$\frac{1}{D-ia}\tan ax$?	$-\frac{1}{a}[1 + ie^{iax}\log(\sec ax + \tan ax)]$.
9	PI of $(D^2+a^2)y = \tan ax$?	$-\frac{1}{a^2}\cos ax\log(\sec ax + \tan ax)$.
10	Other method for $\tan$, $\sec$, $\csc$?	Variation of parameters (class 011; CW P23, P26, P28).
11	Substitution for $X = e^{e^x}$, $\sin(e^x)$?	$t = e^x$, $dt = e^x dx$.
12	$\int te^t\,dt$?	$(t-1)e^t$.
13	ESA Dec-23 3c: solve $y'' + 3y' + 2y = e^{e^x}$.	$y = c_1e^{-x} + c_2e^{-2x} + e^{-2x}e^{e^x}$.
14	$\frac{1}{(D+1)(D+2)}\sin(e^x)$?	$-e^{-2x}\sin(e^x)$.
15	$\frac{1}{(D+2)(D+3)}e^{2x}\sin(e^x)$?	$-\sin(e^x) - 6e^{-x}\cos(e^x) + 18e^{-2x}\sin(e^x) + 24e^{-3x}\cos(e^x)$.
16	Slide slip in this deck?	009·16 line (1): $-e^{-x}\cos(e^x)$ should be $-e^{x}\cos(e^x)$ (final answer right).

Mnemonic (study aid). *"Root outside, minus root inside."* $\frac{1}{D-m}X = e^{+mx}\int Xe^{-mx}dx$ (study aid).

4 Last 10 minutes before the exam

Top facts

1. Not a standard form? Factorise $f(D)$, partial fractions, then $\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}dx$ for each factor; CF unchanged.
2. $\frac{1}{D+1}X = e^{-x}\int Xe^{x}dx$, $\frac{1}{D+2}X = e^{-2x}\int Xe^{2x}dx$; no constants of integration.
3. $\frac{1}{D^2}\ln x = \frac{x^2\ln x}{2} - \frac{3x^2}{4}$.
4. $\frac{1}{D^2+a^2}\tan ax = -\frac{1}{a^2}\cos ax\log(\sec ax + \tan ax)$ (via $\frac{1}{2ia}[\frac{1}{D-ia} - \frac{1}{D+ia}]$, $i \to -i$).
5. ESA Dec-23 3c: $y'' + 3y' + 2y = e^{e^x}$: $y = c_1e^{-x} + c_2e^{-2x} + e^{-2x}e^{e^x}$ $(t = e^x)$.
6. $\frac{\sin(e^x)}{(D+1)(D+2)} = -e^{-2x}\sin(e^x)$; one factor at a time: $\frac{1}{D+2}[-e^{-x}\cos(e^x)]$.
7. $\tan$, $\sec$, $\csc$ can also go by variation of parameters (class 011) — use it when the question names it.

Top formulas

E1.1 · The standard result for one linear factor

ESA ★★

$$\frac{1}{D-m}\,X = e^{mx}\int X\,e^{-mx}\,dx$$

GIVES	The PI contribution of one factor $D-m$.
APPLIES	Any X; m real or complex.
NOTE	Sign: $\frac{1}{D+1}X = e^{-x}\int Xe^{x}dx$ $(m=-1)$. No constant of integration.

S009·4 · T009 00:02:11 · T009b 00:03:01 · T009b 00:03:36 · ESA Dec-23 3c

E1.3 · Partial fractions of $\frac{1}{f(D)}$

ESA ★★

$$\frac{1}{f(D)} = \frac{A_1}{D-m_1} + \cdots + \frac{A_n}{D-m_n}$$

$$\text{P.I.} = A_1e^{m_1x}\int Xe^{-m_1x}dx + \cdots + A_ne^{m_nx}\int Xe^{-m_nx}dx$$

GIVES	The PI as a sum of n single-factor terms.
APPLIES	Distinct roots $m_1, \ldots, m_n$.
NOTE	Cover-up: $A_k = \frac{1}{f'(m_k)}$; e.g. $\frac{1}{(D+1)(D+2)} = \frac{1}{D+1} - \frac{1}{D+2}$.

S009·4 · S009·10 · S009·12 · S009·14 · T009 00:01:32 · T009 00:01:52

E2.5 · Result: PI of $\tan ax$ ESA ★★

$$\frac{1}{D^2+a^2}\tan ax = -\frac{1}{a^2}\cos ax\,\log(\sec ax+\tan ax)$$

GIVES The PI; with the CF, $y = c_1\cos ax + c_2\sin ax - \frac{1}{a^2}\cos ax\log(\sec ax+\tan ax)$.

APPLIES Any $a \neq 0$.

NOTE $a = 2$ gives CW P28's answer (asked there by variation of parameters).

S009·7 · T009 00:11:09 · CW U2 P28

E3.1 · The substitution $t = e^x$ ESA ★★

$$e^{-mx}\int e^{kx}F(e^x)\,dx = e^{-mx}\int t^{k-1}F(t)\,dt, \qquad t = e^x,\ dt = e^x dx$$

GIVES Integrals of powers of e^x times a function of e^x.

APPLIES $X = e^{e^x}$, $\sin(e^x)$, $e^{2x}\sin(e^x)$, ...

NOTE Write $e^{kx}dx = e^{(k-1)x}\cdot e^x dx = t^{k-1}dt$; substitute back $t = e^x$ at the end.

S009·11 · S009·12 · S009·13 · S009·15 · T009 00:14:05 · T009b 00:03:57 · ESA Dec-23 3c

E3.3 · ESA Dec-23 3c: PI of e^{e^x} ESA ★★

$$\frac{1}{D^2+3D+2}\,e^{e^x} = e^{-2x}e^{e^x}$$

GIVES The PI; $y = c_1e^{-x} + c_2e^{-2x} + e^{-2x}e^{e^x}$.

APPLIES $y'' + 3y' + 2y = e^{e^x}$.

NOTE Same as slide 009·10–11; checked by substitution (sympy).

S009·10 · S009·11 · ESA Dec-23 3c